

When Does Aggregating Explanations Work?

PathMNIST Results

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This report consolidates the released PathMNIST experiment results. Tables are grouped by model, experimental protocol, and aggregation setting. Metric directions and robustness definitions are stated in every table header and caption.

Protocol	Model	Settings
Generalization protocol	DenseNet-121	NAIVE, NOISE-S, NOISE-K
Full-matrix protocol	ViT-B/16	NAIVE, IND, NOISE-S, NOISE-K

Table 1: NAIVE results for PathMNIST with DenseNet-121 (Generalization protocol). The methods are the best individual explanation, Simple Averaging, Borda, Kemeny–Young, RRF, and Schulze. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized. The test split contains $n = 7180$ samples.

Method	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
	F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _F ^g $\uparrow$	R _C ^g $\uparrow$	R _C ^g $\uparrow$	R _F ^p $\uparrow$	R _F ^p $\uparrow$	R _C ^p $\uparrow$	R _C ^p $\uparrow$	R _F ^s $\uparrow$	R _F ^s $\uparrow$	R _C ^s $\uparrow$	R _C ^s $\uparrow$	R _F ^a $\uparrow$	R _F ^a $\uparrow$	R _C ^a $\uparrow$	R _C ^a $\uparrow$
Best Individual	0.022	0.756	0.075	0.810	0.008	0.634	0.135	0.249	0.083	0.533	0.222	0.245	0.112	0.462	0.256	0.046	-0.015	0.003	-0.067	0.004
Simple Averaging	0.013	0.816	0.060	0.871	0.028	0.694	0.178	0.310	0.094	0.593	0.249	0.307	0.123	0.522	0.285	0.107	-0.009	0.019	-0.052	0.020
Borda	0.017	0.809	0.064	0.864	0.020	0.687	0.154	0.303	0.097	0.586	0.236	0.299	0.134	0.515	0.275	0.100	-0.014	0.001	-0.058	0.001
Kemeny–Young	0.015	0.812	0.057	0.867	0.021	0.690	0.159	0.306	0.100	0.589	0.241	0.302	0.136	0.518	0.276	0.103	-0.012	0.004	-0.050	0.004
RRF	0.016	0.808	0.061	0.863	0.022	0.686	0.159	0.302	0.098	0.585	0.241	0.298	0.136	0.514	0.276	0.099	-0.013	0.002	-0.054	0.003
Schulze	0.017	0.809	0.062	0.865	0.019	0.687	0.154	0.303	0.097	0.586	0.238	0.300	0.133	0.515	0.272	0.100	-0.014	0.001	-0.055	0.002

Table 2: NOISE-S results for PathMNIST with DenseNet-121 (Generalization protocol), using the Spearman-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _{$\bar{F}$} ^g $\uparrow$	R _C ^g $\uparrow$	R _{$\bar{C}$} ^g $\uparrow$	R _F ^p $\uparrow$	R _{$\bar{F}$} ^p $\uparrow$	R _C ^p $\uparrow$	R _{$\bar{C}$} ^p $\uparrow$	R _F ^s $\uparrow$	R _{$\bar{F}$} ^s $\uparrow$	R _C ^s $\uparrow$	R _{$\bar{C}$} ^s $\uparrow$	R _F ^a $\uparrow$	R _{$\bar{F}$} ^a $\uparrow$	R _C ^a $\uparrow$	R _{$\bar{C}$} ^a $\uparrow$
Simple Averaging	3	0.020	0.797	0.057	0.852	0.018	0.675	0.185	0.291	0.083	0.574	0.249	0.287	0.121	0.503	0.288	0.087	-0.015	0.021	-0.047	0.022
Borda	3	0.027	0.774	0.080	0.828	0.011	0.652	0.157	0.267	0.090	0.551	0.230	0.264	0.133	0.480	0.268	0.064	-0.021	0.030	-0.070	0.031
Kemeny–Young	3	0.029	0.762	0.075	0.816	0.011	0.640	0.154	0.255	0.086	0.539	0.228	0.252	0.127	0.468	0.266	0.052	-0.023	0.026	-0.066	0.027
RRF	3	0.028	0.765	0.077	0.819	0.011	0.643	0.156	0.257	0.090	0.542	0.232	0.254	0.130	0.471	0.269	0.054	-0.022	0.031	-0.067	0.032
Schulze	3	0.027	0.774	0.080	0.828	0.011	0.652	0.157	0.267	0.090	0.551	0.230	0.263	0.134	0.480	0.268	0.064	-0.021	0.030	-0.070	0.031

Table 3: NOISE-K results for PathMNIST with DenseNet-121 (Generalization protocol), using the Kendall-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _{$\bar{F}$} ^g $\uparrow$	R _C ^g $\uparrow$	R _{$\bar{C}$} ^g $\uparrow$	R _F ^p $\uparrow$	R _{$\bar{F}$} ^p $\uparrow$	R _C ^p $\uparrow$	R _{$\bar{C}$} ^p $\uparrow$	R _F ^s $\uparrow$	R _{$\bar{F}$} ^s $\uparrow$	R _C ^s $\uparrow$	R _{$\bar{C}$} ^s $\uparrow$	R _F ^a $\uparrow$	R _{$\bar{F}$} ^a $\uparrow$	R _C ^a $\uparrow$	R _{$\bar{C}$} ^a $\uparrow$
Simple Averaging	3	0.020	0.797	0.057	0.852	0.018	0.675	0.185	0.291	0.083	0.574	0.249	0.287	0.121	0.503	0.288	0.087	-0.015	0.021	-0.047	0.022
Borda	3	0.027	0.774	0.080	0.828	0.011	0.652	0.157	0.267	0.090	0.551	0.230	0.264	0.133	0.480	0.268	0.064	-0.021	0.030	-0.070	0.031
Kemeny–Young	3	0.029	0.762	0.075	0.816	0.011	0.640	0.154	0.255	0.086	0.539	0.228	0.252	0.127	0.468	0.266	0.052	-0.023	0.026	-0.066	0.027
RRF	3	0.028	0.765	0.077	0.819	0.011	0.643	0.156	0.257	0.090	0.542	0.232	0.254	0.130	0.471	0.269	0.054	-0.022	0.031	-0.067	0.032
Schulze	3	0.027	0.774	0.080	0.828	0.011	0.652	0.157	0.267	0.090	0.551	0.230	0.263	0.134	0.480	0.268	0.064	-0.021	0.030	-0.070	0.031

Table 4: NAIVE results for PathMNIST with ViT-B/16 (Full-matrix protocol). The methods are the best individual explanation, Simple Averaging, Borda, Kemeny–Young, RRF, and Schulze. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized. The test split contains $n = 7180$ samples.

Method	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
	F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _{$\bar{F}$} ^g $\uparrow$	R _C ^g $\uparrow$	R _{$\bar{C}$} ^g $\uparrow$	R _F ^p $\uparrow$	R _{$\bar{F}$} ^p $\uparrow$	R _C ^p $\uparrow$	R _{$\bar{C}$} ^p $\uparrow$	R _F ^s $\uparrow$	R _{$\bar{F}$} ^s $\uparrow$	R _C ^s $\uparrow$	R _{$\bar{C}$} ^s $\uparrow$	R _F ^a $\uparrow$	R _{$\bar{F}$} ^a $\uparrow$	R _C ^a $\uparrow$	R _{$\bar{C}$} ^a $\uparrow$
Best Individual	0.186	0.633	0.236	0.681	-0.083	0.508	0.117	0.073	-0.008	0.417	0.183	0.062	0.136	0.243	0.235	0.064	0.137	0.004	0.143	0.003
Simple Averaging	0.139	0.662	0.183	0.715	-0.046	0.531	0.155	0.102	0.030	0.436	0.220	0.093	0.174	0.256	0.284	0.104	0.079	-0.033	0.081	-0.033
Borda	0.133	0.695	0.176	0.748	-0.045	0.562	0.166	0.134	0.032	0.467	0.231	0.126	0.176	0.288	0.286	0.136	0.104	-0.005	0.108	-0.004
Kemeny–Young	0.136	0.698	0.178	0.753	-0.047	0.566	0.177	0.139	0.028	0.473	0.234	0.131	0.170	0.293	0.283	0.141	0.099	-0.005	0.105	-0.005
RRF	0.141	0.694	0.184	0.747	-0.051	0.563	0.168	0.135	0.025	0.469	0.227	0.126	0.167	0.289	0.279	0.136	0.093	-0.006	0.097	-0.005
Schulze	0.135	0.694	0.178	0.748	-0.046	0.561	0.173	0.134	0.030	0.467	0.232	0.126	0.173	0.288	0.284	0.136	0.102	-0.006	0.107	-0.004

Table 5: IND results for PathMNIST with ViT-B/16 (Full-matrix protocol). The Best Individual row is followed by matched NAIVE and IND results for each aggregation method at $q = 11$. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized. The test split contains $n = 7180$ samples.

Method	Setting	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _{$\bar{F}$} ^g $\uparrow$	R _C ^g $\uparrow$	R _{$\bar{C}$} ^g $\uparrow$	R _F ^p $\uparrow$	R _{$\bar{F}$} ^p $\uparrow$	R _C ^p $\uparrow$	R _{$\bar{C}$} ^p $\uparrow$	R _F ^s $\uparrow$	R _{$\bar{F}$} ^s $\uparrow$	R _C ^s $\uparrow$	R _{$\bar{C}$} ^s $\uparrow$	R _F ^a $\uparrow$	R _{$\bar{F}$} ^a $\uparrow$	R _C ^a $\uparrow$	R _{$\bar{C}$} ^a $\uparrow$
Best Individual	baseline	0.110	0.679	0.150	0.728	-0.030	0.542	0.179	0.110	0.047	0.445	0.244	0.098	0.183	0.273	0.299	0.115	0.067	-0.022	0.067	-0.022
Simple Averaging	matched NAIVE	0.100	0.695	0.138	0.744	-0.023	0.559	0.218	0.126	0.053	0.461	0.268	0.116	0.187	0.286	0.308	0.131	0.064	-0.013	0.065	-0.014
Simple Averaging	IND	0.110	0.690	0.148	0.737	-0.027	0.555	0.207	0.121	0.046	0.457	0.255	0.110	0.184	0.283	0.308	0.124	0.068	-0.014	0.068	-0.016
Borda	matched NAIVE	0.099	0.694	0.138	0.743	-0.021	0.558	0.220	0.126	0.056	0.460	0.269	0.115	0.187	0.287	0.305	0.131	0.063	-0.014	0.063	-0.014
Borda	IND	0.109	0.688	0.147	0.737	-0.027	0.553	0.206	0.119	0.048	0.455	0.258	0.108	0.182	0.284	0.303	0.127	0.064	-0.019	0.065	-0.019
Kemeny–Young	matched NAIVE	0.103	0.692	0.140	0.741	-0.026	0.557	0.224	0.124	0.051	0.460	0.269	0.114	0.182	0.286	0.303	0.129	0.065	-0.014	0.065	-0.015
Kemeny–Young	IND	0.110	0.687	0.148	0.736	-0.028	0.552	0.208	0.119	0.046	0.455	0.256	0.108	0.181	0.285	0.301	0.125	0.066	-0.019	0.067	-0.020
RRF	matched NAIVE	0.104	0.690	0.142	0.740	-0.027	0.555	0.220	0.122	0.049	0.458	0.264	0.112	0.181	0.285	0.302	0.129	0.062	-0.016	0.063	-0.017
RRF	IND	0.111	0.687	0.149	0.735	-0.029	0.551	0.208	0.118	0.045	0.455	0.256	0.108	0.179	0.284	0.300	0.125	0.065	-0.018	0.066	-0.019
Schulze	matched NAIVE	0.104	0.692	0.142	0.741	-0.028	0.556	0.220	0.123	0.050	0.458	0.265	0.113	0.181	0.286	0.302	0.130	0.063	-0.014	0.064	-0.016
Schulze	IND	0.112	0.687	0.151	0.736	-0.029	0.552	0.204	0.118	0.045	0.455	0.253	0.108	0.179	0.283	0.298	0.125	0.062	-0.018	0.063	-0.019

Table 6: NOISE-S results for PathMNIST with ViT-B/16 (Full-matrix protocol), using the Spearman-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _{$\bar{F}$} ^g $\uparrow$	R _C ^g $\uparrow$	R _{$\bar{C}$} ^g $\uparrow$	R _F ^p $\uparrow$	R _{$\bar{F}$} ^p $\uparrow$	R _C ^p $\uparrow$	R _{$\bar{C}$} ^p $\uparrow$	R _F ^s $\uparrow$	R _{$\bar{F}$} ^s $\uparrow$	R _C ^s $\uparrow$	R _{$\bar{C}$} ^s $\uparrow$	R _F ^a $\uparrow$	R _{$\bar{F}$} ^a $\uparrow$	R _C ^a $\uparrow$	R _{$\bar{C}$} ^a $\uparrow$
Simple Averaging	2	0.181	0.636	0.230	0.684	-0.081	0.507	0.118	0.073	-0.007	0.415	0.181	0.063	0.135	0.240	0.235	0.066	0.126	-0.002	0.131	-0.003
Borda	2	0.180	0.636	0.228	0.685	-0.079	0.507	0.113	0.073	-0.005	0.413	0.177	0.063	0.137	0.238	0.236	0.064	0.127	-0.001	0.131	-0.001
Kemeny–Young	2	0.180	0.636	0.228	0.685	-0.079	0.507	0.113	0.073	-0.005	0.413	0.177	0.063	0.137	0.238	0.236	0.064	0.127	-0.001	0.131	-0.001
RRF	2	0.181	0.633	0.230	0.682	-0.080	0.505	0.113	0.070	-0.006	0.412	0.177	0.061	0.136	0.237	0.234	0.063	0.127	-0.002	0.130	-0.003
Schulze	2	0.180	0.636	0.228	0.685	-0.079	0.507	0.113	0.073	-0.005	0.413	0.177	0.063	0.137	0.238	0.236	0.064	0.127	-0.001	0.131	-0.001

Table 7: NOISE-K results for PathMNIST with ViT-B/16 (Full-matrix protocol), using the Kendall-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _{$\bar{F}$} ^g $\uparrow$	R _C ^g $\uparrow$	R _{$\bar{C}$} ^g $\uparrow$	R _F ^p $\uparrow$	R _{$\bar{F}$} ^p $\uparrow$	R _C ^p $\uparrow$	R _{$\bar{C}$} ^p $\uparrow$	R _F ^s $\uparrow$	R _{$\bar{F}$} ^s $\uparrow$	R _C ^s $\uparrow$	R _{$\bar{C}$} ^s $\uparrow$	R _F ^a $\uparrow$	R _{$\bar{F}$} ^a $\uparrow$	R _C ^a $\uparrow$	R _{$\bar{C}$} ^a $\uparrow$
Simple Averaging	4	0.165	0.627	0.210	0.677	-0.064	0.504	0.128	0.071	0.012	0.410	0.195	0.060	0.156	0.236	0.261	0.069	0.070	-0.046	0.073	-0.048
Borda	4	0.174	0.647	0.220	0.698	-0.076	0.522	0.129	0.091	0.003	0.432	0.192	0.083	0.147	0.257	0.251	0.092	0.096	-0.025	0.101	-0.024
Kemeny–Young	4	0.187	0.636	0.234	0.685	-0.086	0.512	0.120	0.078	-0.008	0.420	0.180	0.068	0.137	0.245	0.239	0.074	0.105	-0.019	0.108	-0.020
RRF	4	0.181	0.642	0.227	0.692	-0.079	0.518	0.128	0.086	-0.003	0.427	0.188	0.077	0.141	0.252	0.243	0.086	0.097	-0.020	0.102	-0.020
Schulze	4	0.185	0.636	0.231	0.686	-0.085	0.511	0.122	0.079	-0.007	0.420	0.182	0.068	0.137	0.246	0.241	0.076	0.101	-0.017	0.107	-0.018